Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Spatially Resolved Stack Diagnostics and Multi-Physics "Digital Twin" for Scale-Up of AEM Water Electrolysis*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		*Knowledge Services*	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	*Hydrogen*	*R&D / Engineering Services*	*Testing / Inspection / Analysis Techniques*
Concept of Proposed Technology	This technology proposal targets quantitative understanding and optimization of spatial current, flow, and thermal fields inside AEM electrolyzer stacks during scale-up from laboratory cells to kW-class operation. The core concept is to combine (i) advanced electrochemical testing and (ii) FEM-based multi-physics simulation with (iii) non-invasive stack diagnostic imaging to reveal inhomogeneities that cannot be detected by average cell voltage alone. The research institute (Fraunhofer IFAM) will establish a measurement–model correlation framework ("digital twin") that predicts pressure loss, flow distribution, local ohmic losses, and heat generation, and then validates these predictions using magnetic field imaging (MFI) and lock-in thermography (LIT) on operating hardware. The Korean SME partner will integrate IFAM-developed electrodes/porous transport layers (PTL) into practical AEM electrolyzer platforms and provide test operation from single cells up to kW-class stacks, enabling systematic validation in representative environments. The resulting toolchain will deliver spatial maps of local current density and heat signatures and connect them to underlying causes			

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	(e.g., flow maldistribution, PTL heterogeneity, contact resistance, shunt currents), enabling engineering countermeasures such as optimized PTL structures, flow-field layouts, and manufacturing tolerances. The technology is applicable to stack design, quality control, scale-up risk reduction, and lifetime improvement in industrial AEM electrolysis systems.
Rationale for Joint R&D Project	AEM electrolysis scale-up is frequently limited not by intrinsic catalyst activity, but by spatial non-uniformities that emerge at larger active areas and multi-cell stacks: local reactant depletion, bubble accumulation, membrane dehydration/flooding, uneven compression, contact resistance gradients, and temperature hot-spots. These effects cause localized overpotential and accelerated degradation, yet they remain “hidden” when only global stack voltage, average efficiency, or outlet gas composition is measured. As a result, design decisions are often based on trial-and-error iterations, extending development time and increasing technical risk. Joint R&D is required because the research institute contributes specialized expertise and infrastructure in electrochemical characterization, multi-physics FEM simulation (fluid/thermal/electrochemical coupling), and advanced imaging methods (MFI, LIT), while the Korean SME partner provides realistic AEM electrolyzer hardware, scalable cell/stack formats, and long-duration operational testing needed to validate models and diagnostics under representative conditions. The collaboration enables a closed-loop “measure → model → redesign” workflow to (i) identify root causes of spatial inhomogeneity, (ii) quantify sensitivity to design and manufacturing parameters, and (iii) implement engineering solutions with measurable improvements in uniformity, efficiency, and durability. This creates a strong bridge between scientific understanding and industrial implementation, accelerating readiness for commercialization and reducing scale-up uncertainty.
Key Words	1. AEM water electrolysis 2. Magnetic field imaging (local current density 3. Lock-in thermography (thermal mapping) 4. FEM multi-physics simulation / digital twin 5. Stack scale-up and spatial uniformity

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	The project objective is to develop and validate an integrated methodology for spatially resolved performance and degradation analysis in AEM electrolyzer stacks, enabling model-based scale-up. Specifically, the R&D will: (1) establish a standardized electrochemical characterization protocol set (polarization, impedance, transient response, and accelerated stress testing) that is compatible across cell areas and stack formats; (2) build a coupled FEM multi-physics model (electrochemical kinetics + ionic/electronic conduction + two-phase flow/pressure loss + thermal conduction/heat generation) capable of predicting spatial distributions of current density, temperature, and pressure drop; (3) implement non-invasive diagnostic imaging (MFI and LIT) to measure spatial current density and heat maps during operation and to detect early signatures of maldistribution and local degradation; and (4) demonstrate, on a dedicated ~5 kW-class stack validation platform, that spatial non-uniformities can be quantified, attributed to physical causes, and mitigated through design/operational adjustments (e.g., PTL structure, flow-field geometry, compression strategy, and operating parameters). The final outcome will be a validated “digital twin + diagnostics” workflow that delivers design rules and acceptance criteria for uniform operation and improved durability, supporting industrial deployment and providing measurable reduction of scale-up risk.			
R&D Target Specifications	Quantitative targets are defined to ensure technical rigor and measurable progress. Spatial diagnostic capability: produce 2D current density maps via MFI with spatial resolution ≤ 10 mm (or finer if feasible) and repeatability ≤ ±5% under steady operation; produce LIT-based thermal maps with hotspot detection sensitivity of ≤0.2°C (stack surface equivalent) and spatial resolution consistent with stack geometry. Model validation targets: multi-physics model predicts total pressure drop across flow distribution components with deviation ≤ 15% versus measurement; predicts stack-average cell voltage within ±30 mV/cell for defined operating points; and reproduces key spatial features (location and magnitude of maxima/minima in current density) with correlation coefficient R ≥ 0.8 between simulated and measured current-density maps. Uniformity improvement targets (demonstration): reduce coefficient of variation (CV) of local current density across the active area by at least 30% relative to baseline (e.g., CV from ~20% to ≤14%) at a representative operating current density (to be defined based on stack rating, typically ≥1 $A \cdot cm^{-2}$). Durability targets (stack-level): demonstrate stable operation of the 5 kW stack for ≥1,000 h (or an agreed staged durability target) with voltage degradation ≤ 50 mV/1,000 h at a fixed current density point, while showing that spatial non-uniformities identified by MFI/LIT correlate with post-test diagnostics and can be mitigated by design/operational updates.			
R&D Plan	Phase I – Pre-study (Month 0–6): Feasibility, baselines, and Phase II blueprint (~½ page paragraph) Phase I will define the technical baseline and lock in the validation strategy with minimal development risk. The research institute will (i) establish a harmonized test matrix spanning single-cell (≤25 cm^2) and short-stack conditions, including defined operating windows (temperature, current density,			

electrolyte management, and pressure conditions as applicable); (ii) perform baseline electrochemical characterization on reference components to derive initial kinetic/ohmic/mass-transport parameters; and (iii) set up a first-generation multi-physics modeling framework including porous-media flow and thermal conduction to identify dominant drivers of spatial non-uniformity and sensitivity to design parameters (PTL permeability, contact resistance distributions, channel geometry, compression). In parallel, feasibility of MFI and LIT integration will be demonstrated on representative cell/short-stack hardware: mechanical integration concept, electromagnetic/thermal boundary conditions, calibration approach, and data acquisition chain. Deliverables of Phase I are: D-I1 Baseline electrochemical dataset and operating definition, D-I2 initial FEM model with sensitivity study, D-I3 diagnostic integration concept and calibration plan, and D-I4 a detailed Phase II implementation plan with quantified success metrics, risk register, and a design-of-experiments (DoE) plan for optimization loops. A key milestone is M-I: “Go/no-go readiness for Phase II” based on successful demonstration of stable mapping signals (MFI/LIT) and model feasibility on a short-stack surrogate.

Phase II – Main R&D (Month 7–36): Development, validation, and 5 kW stack demonstration

WP2.1 – Electrochemical test protocol development and parameter identification (M7–M14) (~½ page paragraph)

This work package standardizes electrochemical diagnostics so that results remain comparable across hardware scales and operating points. The institute will develop a protocol suite covering polarization curves, EIS over operating current density, transient load steps, and accelerated stress tests (AST) tailored to AEM electrolysis. The aim is to deconvolute voltage losses into kinetic, ohmic, and mass-transport contributions and to extract model parameters (exchange current densities, effective conductivity, contact resistance, mass-transport limiting behavior) with quantified uncertainty. The Korean SME partner will provide cell/stack operation and facilitate repeatable test execution on defined hardware configurations. Deliverables include D2.1 validated electrochemical protocol handbook, D2.2 parameter set with uncertainty bounds for modeling, and D2.3 degradation signatures library linking electrochemical features (e.g., EIS arc evolution) to likely physical drivers. Milestone M2.1 is achieved when parameter sets reproduce measured polarization/EIS within predefined tolerances at multiple operating points and when protocol repeatability is demonstrated (e.g., ≤±10 mV variation in repeated polarization at a reference point).

WP2.2 – FEM multi-physics digital twin for spatial current/flow/thermal fields (M7–M20) (~½ page paragraph)

WP2.2 develops a coupled simulation toolchain that scales from cell to stack. The institute will implement FEM models integrating (i) porous media flow through PTL and channels (including pressure loss and distribution), (ii) thermal conduction and heat generation (ohmic + overpotential contributions), and (iii) electrochemical current distribution including ionic/electronic conduction pathways and contact resistance effects. Where necessary, reduced-order models will be derived to enable fast sensitivity screening, while high-fidelity simulations will be retained for final validation. A

specific objective is scaling logic: translating material-level properties (PTL permeability, compressibility, thermal conductivity, effective conductivity) into stack-level distributions and performance impact. Validation interfaces will be defined such that simulation outputs can be compared directly with MFI-derived current maps and LIT-derived thermal maps. Deliverables: D2.4 digital twin model v1 (cell/short-stack), D2.5 validated pressure drop and thermal prediction module, and D2.6 scaling guideline document identifying which material and geometry parameters dominate non-uniformity. Milestone M2.2 is reached when predicted pressure drops and temperature profiles match experimental values within target deviations and when the model produces a spatial current distribution hypothesis that can be tested by imaging.

WP2.3 – Magnetic field imaging for local current density mapping (M10–M26) (~½ page paragraph)
This work package develops a robust MFI methodology for operating AEM stacks. The institute will design the sensor arrangement (e.g., multi-axis magnetic sensors) and data acquisition to capture the magnetic field around the stack under realistic currents. A key scientific task is the inverse reconstruction problem: converting magnetic field measurements into current density maps using physical constraints (Biot–Savart based forward models, regularization, boundary conditions) and calibrating the reconstruction with known reference states and/or segmented measurements where feasible. Signal disturbance sources (external fields, cabling, busbars, stray currents) will be modeled and compensated. The Korean SME partner supports mechanical integration and stable operation to enable repeated measurements. Deliverables: D2.7 MFI hardware + acquisition workflow, D2.8 reconstruction algorithm and validation report (accuracy, repeatability), and D2.9 current density mapping results on a representative short-stack configuration. Milestone M2.3: demonstration of repeatable spatial mapping under at least three operating points and documented correlation of map features with controlled perturbations (e.g., induced flow restriction or controlled contact resistance change).

WP2.4 – Lock-in thermography for spatial thermal diagnostics and hotspot detection (M10–M26) (~½ page paragraph)
WP2.4 adapts lock-in thermography (LIT) to AEM electrolyzer stack environments for non-invasive identification of localized heat generation and thermal transport anomalies. The institute will define modulation strategies (current or load modulation, frequency selection) to separate periodic thermal signals from background drift and to improve sensitivity to localized losses. Thermal boundary conditions, emissivity calibration, and the influence of stack insulation/housing will be quantified to allow meaningful spatial interpretation. The goal is not merely surface imaging, but deriving actionable thermal indicators (hotspot intensity, phase shift, thermal diffusion patterns) that relate to internal phenomena such as increased ohmic resistance, poor reactant distribution, or local overpotential. Deliverables: D2.10 LIT measurement protocol + calibration, D2.11 thermal hotspot catalog and thresholds, and D2.12 cross-correlation report linking thermal features to electrochemical metrics (e.g., EIS changes) and to modeled heat sources from WP2.2. Milestone M2.4 is achieved when LIT can reproducibly detect intentionally created non-uniformities and when hotspot locations show

consistent alignment with MFI-identified current density extremes.

WP2.5 – Integrated stack-level validation on ~5 kW platform and optimization loops (M18–M34) (~½ page paragraph)

This is the central demonstration WP, integrating electrochemical testing, simulation, and imaging on a ~5 kW-class stack to analyze spatial current and flow fields and to implement mitigation measures. The Korean SME partner provides the stack platform, operational capability, and controlled variation of design/operating parameters. The institute performs synchronized measurement campaigns combining: (i) electrochemical characterization (polarization/EIS and transients), (ii) MFI current mapping, (iii) LIT thermal mapping, and (iv) model predictions from the digital twin. A structured DoE will vary parameters such as flow distribution elements, PTL structures, compression settings, and operating conditions to quantify their impact on spatial uniformity metrics (CV, max/min ratios) and on performance/durability. Iterative optimization loops will be performed where simulation suggests countermeasures that are then experimentally validated, generating engineering rules for homogeneous operation. Deliverables: D2.13 validated "measurement–model" workflow, D2.14 stack spatial uniformity improvement demonstration, and D2.15 design rules and acceptance criteria (e.g., permissible pressure drop ranges, PTL property windows, thermal gradient limits). Milestone M2.5: achievement of the defined uniformity improvement target (relative CV reduction) and model–measurement correlation targets at the 5 kW stack level.

WP2.6 – Durability assessment, degradation localization, and transfer guidelines (M24–M36) (~½ page paragraph)

WP2.6 connects spatial non-uniformities to lifetime and defines practical mitigation and QC guidance for scale-up. Long-term operation and/or staged durability tests will be conducted on the 5 kW stack using defined duty cycles. MFI and LIT will be applied periodically to track the evolution of spatial patterns and to identify early indicators of localized degradation. Post-test analyses (e.g., contact resistance mapping, PTL inspection, membrane/electrode degradation assessment) will be used to close the loop between observed spatial signatures and physical failure modes. The institute will update the digital twin parameters using observed degradation trends (e.g., evolving resistances, transport limitations), enabling predictive lifetime insight. Deliverables: D2.16 spatial degradation atlas (how patterns evolve over time), D2.17 mitigation playbook (operational/design countermeasures), and D2.18 transfer guideline package for implementation in future stacks including recommended instrumentation options and QC thresholds. Milestone M2.6 is achieved when the project demonstrates that spatial diagnostics can (i) detect emerging issues before major performance loss and (ii) guide corrective actions that measurably reduce degradation rate relative to baseline.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute (Fraunhofer IFAM) will lead the scientific and methodological development of the electrochemical diagnostics, multi-physics simulation, and non-invasive imaging toolchain. Responsibilities include: definition and execution of standardized electrochemical test protocols; extraction of physically meaningful model parameters with uncertainty quantification; development and validation of FEM-based coupled models for flow, pressure loss, thermal transport, and current distribution; implementation and calibration of MFI and LIT measurement setups; reconstruction algorithms for deriving local current density distributions; and correlation of measurement results with simulation to establish a validated digital twin workflow. The institute will also lead the interpretation of spatial non-uniformities, identify root causes, propose mitigation measures, and document design rules and acceptance criteria. Quality management includes defining KPIs for model accuracy, mapping repeatability, and improvement targets, and ensuring traceability of data and modeling assumptions across cell and stack scales.
Expected R&R of Korean SME Partner	The Korean SME partner will provide the practical AEM electrolyzer platforms required for representative validation, including single-cell, short-stack, and kW-class stack operation capability, and will integrate research-institute electrode/PTL concepts into these platforms as appropriate. The SME partner will be responsible for stack assembly, safe operation, and stable test execution under defined operating windows, including provision of process control data (flows, pressures, temperatures, power input, gas output) and long-term durability operation. The partner will support mechanical/electrical integration of MFI and LIT hardware (mounting concepts, access, shielding considerations) and implement design variations needed for DoE studies (e.g., flow distribution configurations, compression strategies, component tolerances). The SME partner will also contribute system-level constraints and commercialization requirements to ensure that resulting guidelines are actionable for industrial deployment.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project will deliver a validated and transferable methodology to measure, model, and improve spatial uniformity in AEM electrolyzer stacks, addressing a major barrier in scale-up and durability. Technically, the expected outcomes include (i) a validated electrochemical protocol library and parameter database for AEM operation; (ii) a coupled FEM multi-physics digital twin capable of predicting pressure drop, thermal behavior, and spatial current density distributions; (iii) demonstrated non-invasive stack diagnostics using magnetic field imaging and lock-in thermography with quantified accuracy and repeatability; and (iv) a 5 kW-class stack demonstration showing measurable reduction of current-density non-uniformity and an associated improvement in stability and degradation rate. The industrial impact is reduced development time and lower scale-up risk through early detection of maldistribution and hotspots, enabling design corrections before costly large-scale iterations. The resulting design rules and acceptance criteria support more robust stack architectures, improved lifetime, and enhanced efficiency, strengthening competitiveness of AEM electrolyzer technology in hydrogen production markets. The approach also creates a foundation for future “smart stack” monitoring concepts and accelerates technology transfer from applied research into deployable industrial products.

6. References

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(2)

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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